



Understanding particle margination in blood flow – A step toward optimized drug delivery systems

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ABSTRACT

Targeted delivery of drugs and imaging agents is very promising to develop new strategies for the treatment of various diseases such as cancer. For an efficient targeted adhesion, the particles have to migrate toward the walls in blood flow – a process referred to as margination. Due to a huge diversity of available carriers, a good understanding of their margination properties in blood flow depending on various flow conditions and particle properties is required. We employ a particle-based mesoscopic hydrodynamic simulation approach to investigate the margination of different carriers for a wide range of hematocrits (volume fraction of red blood cells) and flow rates. Our results show that margination strongly depends on the thickness of the available free space close to the wall, the so-called red blood cell-free layer (RBC-FL), in comparison to the carrier size. The carriers with a few micrometers in size are comparable with the RBC-FL thickness and marginate better than their sub-micrometer counterparts. Deformable carriers, in general, show worse margination properties than rigid particles. Particle margination is also found to be most pronounced in small channels with a characteristic size comparable to blood capillaries. Finally, different margination mechanisms are discussed.

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1. Introduction

One of the major causes of human death in the world is cancer; for instance, about 8 million people died from cancer in 2012 [1]. Although significant progress has been achieved in developing treatments for cancer, the impact on patient survival remains rather moderate [2]. Therefore, the development of early detection and therapy strategies for cancer is under active research. One of the ideas with a great potential in this area is the targeted delivery of drugs and imaging agents through micro- and nano-particles [2]. A very challenging task is to design suitable carriers which would meet several demands including good adhesion properties to targeted sites in blood flow, efficient transport through the biological barriers (e.g., vessel walls, interstitial space, and cell membranes), and low clearance by various defense mechanisms of the body [3–7]. Proposed solutions are polymer conjugates, which are already in clinical use [8,9], fabricated nano-particles [9,10], and self-assembled structures from lipids or block copolymers forming liposomes, polymersomes, or worm-like micelles [9]. All these micro- and nano-carriers differ in shape, size, and deformability. Furthermore, the circulatory system of humans consists of blood vessels which vary from several centimeters to a few micrometers with a wide range of flow rates. Hence, to identify

advantages and disadvantages of various particles for vascular drug delivery, their behavior in blood flow for different flow rates, hematocrits (i.e., volume fraction of red blood cells), vessel diameters, particle sizes, shapes, and deformability needs to be better understood.

Blood is a complex fluid which consists of red blood cells (RBCs), white blood cells (WBCs), platelets, and the blood plasma. The physiological systemic hematocrit is in the range 37%–54%, varying for males and females [11]. The local hematocrit in microcirculation can be even lower and depends on the vessel diameter [12,13]. Experimental [14–19], theoretical [20–22], and numerical [23–26] studies have shown that vesicles and RBCs experience a lift force acting away from a wall due to the hydrodynamic interactions with the wall and the RBC shape and deformability. Consequently, RBCs in blood flow migrate toward the center of a vessel and close to the wall a region depleted of RBCs develops, called the RBC-free layer (RBC-FL). In contrast, WBCs and platelets migrate toward vessel walls, which is referred to as margination [27–30]. The margination of WBCs and platelets is mediated by RBCs and appears to be important for these cells, since in order to perform their biological function they must have a possibility to adhere to the wall. Thus, margination can be thought of as a necessary precondition for the wall adhesion.

In similarity with WBCs and platelets, the margination process in blood flow is also expected to occur for micro- and nano-particles. Experiments [31] on margination of rigid platelet-like particles in channels with a characteristic size within 50 μm –200 μm have shown that a significant number of the particles gets accumulated

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near a wall if the hematocrit is larger than about 7% and the accumulation is enhanced with increasing hematocrit. Recent microfluidic experiments [32] have shown that rigid spheres with the size of 2 μm display a significantly higher adhesion density than particles with a size of 200 nm–500 nm. Thus, available experiments already show that there exists a strong dependence of particle margination on hematocrit, flow rate, and carrier size. Numerical simulations of blood flow are ideally suited to explore the margination trends for a wide range of conditions and to better understand the corresponding mechanisms.

Recent numerical simulations have considered segregation of a binary mixture of spherical elastic capsules in shear and channel flow [33–35]. Simulations for different capsule sizes have shown that in binary mixtures of particles, the large particles migrate toward the center and the small ones toward the wall if the volume fraction of the small ones is low. Furthermore, floppy particles tend to migrate to the center in a binary mixture together with rigid particles. Simulations of WBCs in two dimensions (2D) [36,37] and three dimensions (3D) [38] have shown that WBCs are subject to strong margination for $H_t \approx 0.2$ – 0.4 and the venular range of flow rates, characterized by the pseudo-shear rates $\dot{\gamma} \lesssim 80 \text{ s}^{-1}$ in small vessels [39,40]. Platelet margination has been studied in 2D for two shear rates, two hematocrit values ($H_t = 0.2$ and $H_t = 0.4$), and two sizes (0.75 μm and 1.5 μm) of rigid particles [41] in a 50 μm wide channel. The results indicate that platelet margination increases with hematocrit, shear rate, and particle size. A similar trend is observed in another 2D and 3D simulation study of particles between 0.25 μm and 1 μm in a 20 μm wide channel [42], demonstrating that the margination is significantly worse for sub-micrometer particles in comparison to larger carriers in agreement with experimental findings [32]. This numerical investigation has also shown that an oblate discoidal shape has advantages for drug delivery systems compared to spheres. Other 3D simulations [43] have focused on the dependence of platelet margination on hematocrit, their aspect ratio, and different viscosities between the inner and outer fluids of RBCs. The simulation results have shown that margination of rigid particles increases with increasing hematocrit and that the RBC viscosity contrast influences margination as well. Additionally, the study indicated that discs marginate slower than spherical particles. Furthermore, platelet margination in blood flow has been studied numerically for different shear rates and hematocrit values of $H_t = 0.1$ and $H_t = 0.2$ [44,45] for rigid discoidal and spherical particles in a channel of 34 μm . In these investigations, an increase of margination with hematocrit has been observed and related to the different RBC-FL thickness.

In this paper, we systematically investigate the dependence of margination on particle size for a wide range of hematocrit values and flow rates using a 3D model system. In this context, we concentrate on the margination into a thin layer near the wall, where particle adhesion to the wall would be possible. Thus, we assess the adhesion potential of particles in blood flow depending on various carrier and flow properties. In addition, we employ 2D simulations to study the effect of particle deformability on its margination properties. Finally, we consider blood flow and particle margination in channels of several widths. Our results from 3D simulations further support the observations that particles with a few micrometers in size marginate significantly better than their sub-micrometer counterparts. Deformable carriers are in general worse than rigid particles; however, they may marginate slightly better at high H_t values and low shear rates, where their deformability aids them to fit better within a relatively narrow RBC-FL at such conditions. Furthermore, the margination of particles is found to be most efficient in small channels with the sizes corresponding to capillaries in the human microvasculature. Finally, we will also discuss the physical mechanisms of margination and the relation between particle margination, its physical properties, and the thickness of the RBC-FL.

2. Methods and models

2.1. Simulation methods

Two particle-based hydrodynamic simulation approaches, the dissipative particle dynamics (DPD) [46,47] method and the smoothed dissipative particle dynamics method with angular momentum conservation (SDPD+a) [48], have been employed. DPD has mainly been used for 2D simulations and SDPD+a for 3D simulations. An advantage of SDPD is the possibility to directly control the fluid compressibility and viscosity in comparison to DPD, where these quantities need to be pre-calculated in a separate simulation. In both methods, the system is represented by a collection of N_p point particles. These particles interact locally through three pairwise additive forces: a conservative force $\mathbf{F}^C$, a dissipative or drag force $\mathbf{F}^D$, and a random $\mathbf{F}^R$ force. Hence, the total force $\mathbf{f}_i$ on particle i exerted by all neighboring particles j is given by

$$\mathbf{f}_i = \sum_{j \neq i} (\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R). \quad (1)$$

The conservative force $\mathbf{F}^C$ controls the pressure and fluid compressibility, while the pair of dissipative $\mathbf{F}^D$ and random $\mathbf{F}^R$ forces defines a local thermostat in order to keep the system at a constant equilibrium temperature. The forces are local and vanish beyond a selected cut-off radius r_c in DPD or a smoothing length r_h in SDPD.

In DPD, the proposed forces act only along the separation vector $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ of the particles [49–51]. In SDPD, the forces are derived through a discretization of the Navier–Stokes equation as for the smoothed particle hydrodynamics [52] method, while the incorporation of thermal fluctuations is similar to that in DPD [53]. The original SDPD formulation [53] did not satisfy angular momentum conservation, which is crucial in applications where fluids with different viscosities coexist [48]. To incorporate angular momentum conservation, an additional spin variable for every particle ω_i has been introduced [48]. Therefore, the dissipative force $\mathbf{F}^D = \mathbf{F}^{D_L} + \mathbf{F}^{D_R}$ in SDPD+a is modified and consists of two force components, a translational friction $\mathbf{F}^{D_T}$ and a rotational friction $\mathbf{F}^{D_R}$. The time evolution of the position $\mathbf{r}$ and the translational velocity $\mathbf{v}$ in DPD and SDPD+a, as well as the angular velocity ω in SDPD+a, of a particle i follows Newton's second law,

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad \dot{\mathbf{v}}_i = \sum_j \frac{1}{m_j} \mathbf{F}_{ij}, \quad \dot{\omega}_i = \sum_j \frac{1}{I_j} \mathbf{N}_{ij}, \quad (2)$$

where m_j is the mass of particle j , I_j is the moment of inertia, and $\mathbf{N}_{ij} = -\mathbf{r}_{ij} \times \mathbf{F}_{ij}/2$ is the torque exerted by particle j on particle i . The equations of motion for both methods are integrated in time by the velocity-Verlet algorithm [54].

2.2. Model for RBCs and particles

Network models have been successfully used to model membranes of vesicles and RBCs filled with a cytosol [55–58]. Here, we employ a network model [57,58] to simulate both, the membranes of RBCs and drug carriers. In 3D, the network is defined by a 2D triangulated mesh on a membrane surface, see Fig. 1(a). The vertices are connected by N_s springs and form N_t triangles. The full potential energy of the system is given by

$$U = U_{\text{spring}} + U_{\text{bending}} + U_{\text{area}} + U_{\text{volume}}, \quad (3)$$

with a spring potential U_{spring} , a bending potential U_{bending} , and area and volume conservation constraints U_{area} , U_{volume} , respectively. For the RBCs, a membrane viscosity is also employed, which is necessary to mimic realistic rheological properties of an RBC membrane [58]. For model details we refer the reader to [57].

In 3D, RBCs are characterized by the effective diameter $D_r = \sqrt{A_0/\pi}$, where A_0 is the RBC surface area. Typical physiological values

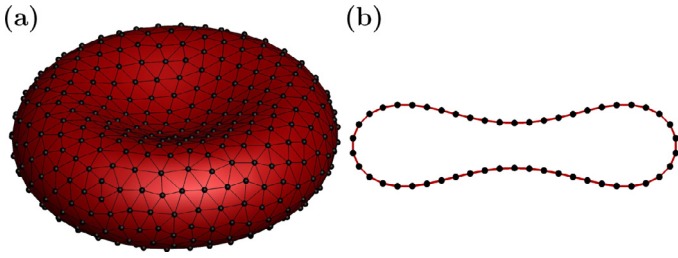


Fig. 1. RBC models. (a) Triangulated network on an RBC surface in 3D. (b) 2D model of an RBC using the bead-spring representation.

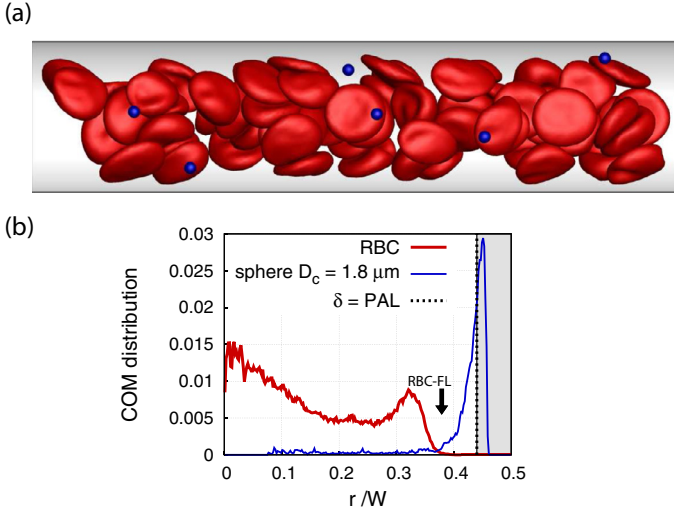


Fig. 2. Distribution of RBCs and carriers in blood flow. (a) A snapshot for the shear rate of $\dot{\gamma}^* \approx 59$ and the hematocrit $H_t = 0.3$ in a tube with the diameter $W = 20 \mu\text{m}$. RBCs are shown in red, while carriers with the diameter $D_c = 0.25 D_r$ in blue. (b) Center-of-mass (COM) probability distributions depending on the radial position r normalized by the diameter $W = 20 \mu\text{m}$ of the channel. The black arrow indicates the RBC-FL and the PAL is shown by the shaded gray area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for healthy blood conditions are $D_r = 6.5 \mu\text{m}$, the plasma viscosity $\eta = 1.2 \times 10^{-3} \text{ Pas}$, and the bending rigidity of an RBC membrane κ_r within the range of $50\text{--}70 k_B T$. RBCs are further characterized by the reduced volume $V^* = 6V_0/(\pi D_r^3) = 0.64$, where V_0 is the enclosed RBC volume.

In 2D, RBCs and particles are modeled as closed bead-spring chains (see Fig. 1(b)) with a similar spring model as in 3D, a bending potential between two neighboring springs, and an area conservation constraint [37]. The RBCs in 2D are characterized by the effective diameter $D_r = L_0/\pi$, where L_0 is the cell contour length. A typical value is $D_r = 6.1 \mu\text{m}$. RBCs are further characterized by the reduced area $A^* = 4A_0/(\pi D_r^2) = 0.46$. The overlapping of different cells is prevented by a truncated Lennard–Jones repulsion between the cell vertices and bounce-back reflections of the vertices of one cell on the triangle surfaces of the other cells. More details on simulation parameters can be found in [42]. More information on RBC and blood flow modeling can be found in recent reviews [58–62].

2.3. Simulation setup

A typical model system corresponds to a cylindrical microvessel in 3D with diameter W and length of $L_c = 12.3 D_r$, as shown in Fig. 2(a) and a slit geometry in 2D with a width W and a length $L_c = 19.5 D_r$, as shown in Fig. 6. The wall boundaries are modeled by frozen fluid particles with the same structure (e.g., the radial distribution function) as that of the fluid. In the flow direction, periodic boundary conditions are assumed and blood flow is driven by a constant force applied

to each solvent particle, which is equivalent to a prescribed pressure drop. In order to characterize the flow strength, a non-dimensional shear rate in both 2D and 3D is defined as

$$\dot{\gamma}^* = \bar{\gamma} \tau_{\text{RBC}} = \bar{\gamma} \frac{\eta D_r^3}{\kappa_r}, \quad (4)$$

where κ_r is the bending rigidity, $\bar{\gamma} = \bar{v}/W$ is the average shear rate (or pseudo shear rate), and $\bar{v}$ is the average flow velocity computed from the velocity profile, while τ_{RBC} defines a characteristic RBC relaxation time. The simulated values of $\dot{\gamma}^*$ cover the range of flow rates characteristic for the venular part of microcirculation ($\bar{\gamma} \lesssim 80 \text{ s}^{-1}$ for $W \approx 20 \mu\text{m}$), where it is estimated that $\dot{\gamma}^* \lesssim 90$ in 3D and $\dot{\gamma}^* \lesssim 104$ in 2D, while in arteriolar part the flow rates are larger ($\bar{\gamma} \gtrsim 110 \text{ s}^{-1}$ for $W \approx 20 \mu\text{m}$) with $\dot{\gamma}^* \gtrsim 120$ in 3D [39,40]. The considered range of shear rates is also relevant for tumor microvasculature, since blood flow velocities in tumors are much reduced in comparison to those under normal conditions, due to large geometric resistance and vessel permeability [63,64].

3. Results

The sampling of cross-sectional center-of-mass (COM) positions of the carriers in blood flow over time leads to a distribution, which reflects the probability of a carrier to be at a certain radial position r . Fig. 2(b) displays a COM distribution for RBCs and carriers with the diameter $D_c = 0.25 D_r$ at $H_t = 0.3$ and $\dot{\gamma}^* \approx 59$. In contrast to RBCs, the carriers migrate into the RBC-FL, a region depleted of RBCs next to the wall, and remain quasi-trapped there. To determine the RBC-FL thickness, the outer edge of the RBC core can be measured, which is similar to the RBC-FL measurements in experiments [65,66]. In order to quantify and compare carrier margination for a wide range of flow and particle parameters, we introduce a margination probability of the carrier's COM to be within a certain distance δ away from the wall. Suitable distances would be the RBC-FL thickness or, for instance, the potential adhesion layer (PAL) defined as the region where adhesive interactions between the carrier's surface and the wall would be possible. We will mainly concentrate on the margination into the PAL in this study.

The margination probabilities p_δ are calculated in 3D (or 2D) by integrating the COM distributions P_{COM} over a certain region as $p_\delta = 2\pi \int_{W/2-\delta}^{W/2} r P_{\text{COM}}(r) dr$ ($p_\delta = 2 \int_0^\delta P_{\text{COM}}(y) dy$). The COM distributions are normalized, such that $p_\delta = 1$ for $\delta = W/2$. The analysis of margination into the PAL with $\delta_{\text{PAL}} = D_c/2 + s$ is restricted to a distance $s = 200 \text{ nm}$ between the carrier's surface and the wall, due to resolution restrictions of the applied mesoscopic simulation method. Although the distance s is motivated by the receptor-ligand interactions, which would normally occur within several nanometers, the distance of several hundred nanometers becomes relevant for carrier-wall interactions in the case of a carrier whose surface is decorated by tethered molecules [67], or for vessels decorated with proteins such as the von Willebrand factor [68]. These margination probabilities are then used to construct so-called margination diagrams, in order to compare the margination for a wide range of shear rates and H_t values.

3.1. Margination of carriers of different sizes in 3D

First, we consider margination of spherical particles with three different diameters $D_c \in \{0.3, 0.25, 0.15\} D_r$, which are nearly non-deformable in comparison to RBCs, see Fig. 3(a). The corresponding margination probability diagrams are displayed in Fig. 3(b)–(d) for different sizes of spheres. The diagrams show that the margination increases with increasing hematocrit. In contrast, the margination with respect to shear rate may first increase at low $\dot{\gamma}^*$, but generally worsens at high shear rates. An important finding here is that the margination probabilities decrease as the carrier size gets smaller,

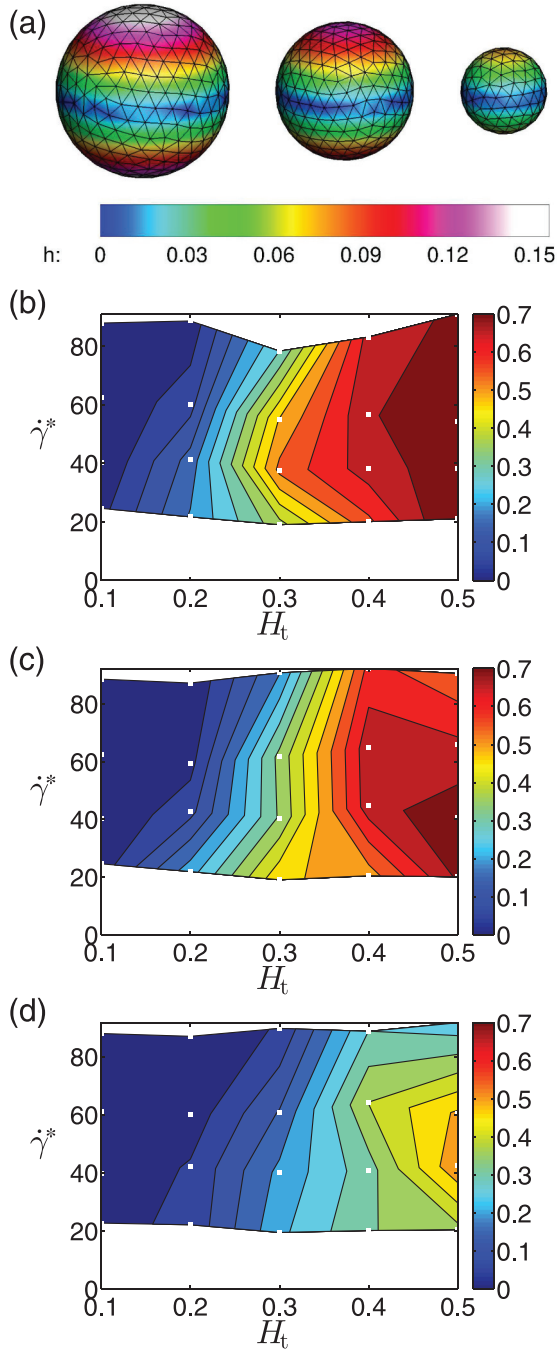


Fig. 3. Carrier margination for different sizes. (a) Schematic of simulated 3D carriers. The absolute value of the z coordinate denoted by h is given by the color code: from blue ($h = 0$) to white ($h = 0.15D_r$), with the RBC diameter D_r . (b, c, d) Probability of margination into the PAL displayed for different hematocrit values H_t and shear rates $\dot{\gamma}^*$. The considered diameters are: (b) $D_c = 0.3D_r$, (c) $D_c = 0.25D_r$, and (d) $D_c = 0.15D_r$. The white squares ($\square$) indicate the values of H_t and $\dot{\gamma}^*$ for which simulations have been performed. The contour plots are obtained using linear interpolation between the simulated points. The color code ranges from blue (low probability) to red (high probability). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

which appears to be consistent with recent microfluidic experiments [32]. Especially, for $D_c = 0.15D_r$ the margination is drastically reduced compared to the other sizes. The differences in margination for the two cases $D_c \in \{0.3, 0.25\}D_r$ appear to be rather small, although the volume of the sphere with $D_c = 0.3D_r$ is twice the volume of the sphere with $D_c = 0.25D_r$. As a conclusion, the characteristic size of a carrier mainly affects its margination properties rather than its

volume. These results are also in excellent qualitative agreement with the investigation of carrier margination in 2D systems [42].

3.2. Effect of particle deformability

The results in Section 3.1 correspond to nearly rigid particles, which is also relevant for the margination of platelets and WBCs in blood flow, since these cells are considerably more rigid than RBCs [44,69]. Even though many drug carriers can be considered rather rigid, several proposed therapeutic particles might be deformable, such as polymer conjugates and liposomes [9]. The deformability is one of the reasons for a particle drift away from the wall due to a hydrodynamic lift force, which is well illustrated by RBCs. Hence, we investigate the influence of particle deformability on its margination properties in blood flow.

In the simulations, a soft 2D particle with $D_c = 0.3D_r$ is modeled similar to an RBC; thus, it has the same parameters for the springs, area constraint and bending coefficients as an RBC. In contrast, a rigid carrier is modeled by a collection of points which are constrained to maintain rigid configuration. Fig. 4(a) and (b) shows the margination diagrams for the rigid (left) and soft (right) carriers. These diagrams indicate that in general the rigid carrier marginates better than the corresponding soft particle, which is most evident at high shear rates. However, at low shear rates their performance is comparable, since shear stresses are not strong enough to significantly deform soft particles. For a more quantitative comparison, Fig. 4(c) presents diagram cuts for the shear rates $\dot{\gamma}^* = 6.1$, $\dot{\gamma}^* = 20.8$, and $\dot{\gamma}^* = 43.4$. Here, the margination probability is plotted against the thickness $\Delta_{\text{RBC-FL}}$ of the RBC-FL. Fig. 4(c) shows that at the lowest shear rate the soft carrier is slightly better marginated than the rigid one. However, with the increasing shear rate the margination of rigid particles becomes clearly better than that of the soft particles. This is due to the increase in deformation of soft particles as the shear rate increases, since stronger deformations lead to an increase of the lift force which drives a particle away from the wall. However, as long as the lift force is small enough a deformable particle is able to come closer to the wall than a rigid carrier (again due to its deformability) providing a better margination into the PAL at low shear rates. A convenient measure for the magnitude of deformation is the asphericity

$$\alpha = \left\langle \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} \right\rangle, \quad (5)$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are the two eigenvalues of the moment-of-inertia tensor in 2D; for a circle $\alpha = 0$ and for a thin rod $\alpha = 1$ [70]. Fig. 5 confirms that the asphericity strongly increases with increasing hematocrit and shear rate. In conclusion, the strong decrease of the margination of soft particles with the increasing shear rate is likely to significantly affect the adhesion characteristics of drug carriers made of polymer conjugates.

3.3. Margination in vessels with different sizes

Our simulation results have shown that particle margination worsens with decreasing particle size. To investigate whether the channel width affects particle margination in blood flow, we employ different channel sizes in simulations, see Fig. 6. Recent simulations [42] have shown that margination of carriers with $D_c = 0.3D_r$ and $D_c = 0.15D_r$ is significantly better in a 10 μm channel in comparison to wider channels with sizes 20 μm and 40 μm . Here, we also investigate a larger particle size of $D_c = 0.6D_r$.

Fig. 7(a) presents the probability p_{PAL} of margination into the PAL for the particle sizes $D_c = 0.3D_r$ and $D_c = 0.6D_r$ and the two channel sizes $W = 1.64D_r$ and $W = 3.28D_r$ with respect to the RBC-FL thickness. The results indicate that for the smallest RBC-FL thickness in the case of $W = 3.28D_r$ with $D_c = 0.6D_r$ as well as $W = 1.64D_r$

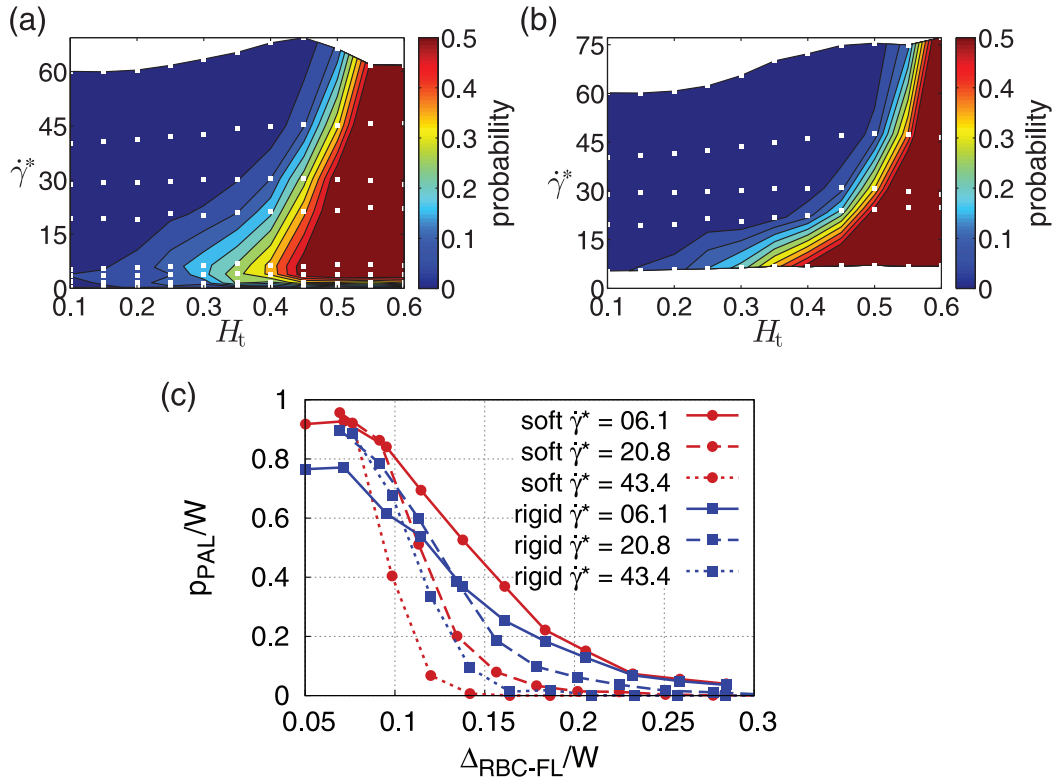


Fig. 4. Comparison of margination of deformable and rigid carriers. (a,b) Probability of margination into the PAL. A circle carrier with $D_c = 0.3D_r$ is constrained to be (a) rigid or (b) deformable as an RBC. The white squares ($\square$) indicate the values of H_t and $\dot{\gamma}^*$ for which simulations have been performed. The contour plots are obtained using linear interpolation between the simulated points. The color code ranges from blue (low probability) to red (high probability). (c) Probability p_{PAL} of margination depending on the RBC-FL thickness Δ_{RBC-FL} and scaled by the channel width W . A comparison of the deformable (red circles) and rigid (blue squares) carriers, whose diameter is $D_c = 0.3D_r$. The shear rates $\dot{\gamma}^* = 6.1$ (solid curves), $\dot{\gamma}^* = 20.8$ (dashed curves), and $\dot{\gamma}^* = 43.4$ (dotted curve) are considered. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with $D_c = 0.3D_r$, the margination probability remains nearly constant or even decreases depending on the shear rate. Consistently, the margination probability for the larger carrier is indeed higher and the margination probability for the smaller channel is also the largest in agreement with the results in [42].

The dependence of particle margination on the channel size can be best understood by looking at the particle distributions. Fig. 7(b) shows that the particle distribution becomes broader as the channel size is increased. Therefore, particles spend less time in the PAL, as illustrated by comparison of the red ($W = 1.64D_r$) and green ($W = 6.56D_r$) curves in Fig. 7(b). However, for an even broader channel ($W = 9.84D_r$) the margination probability into the PAL may also increase, as shown by the green and purple curves. For the largest channel size faster carriers located further away from the wall overtake slower carriers and effectively push them closer to the wall, see Fig. 6(d). This leads to the formation of two main paths of moving carriers supported by the double peak in the COM distribution with a distance of the two local maxima of about a carrier diameter. However, such an effect is mainly expected when carrier concentration is large enough. Finally, this effect does not arise in the smaller channels, since there the RBC-FL thickness is too small to accommodate the two particle diameters.

4. Discussion

Our simulation results clearly support the idea that hematocrit, shear rate, particle size, and channel size strongly affect the efficiency of particle margination. The presence of RBCs is essential in this process, since almost all of the observed effects can be related to the change of the RBC-FL thickness. Without RBCs, which populate the channel center and result in the RBC-FL, suspended particles

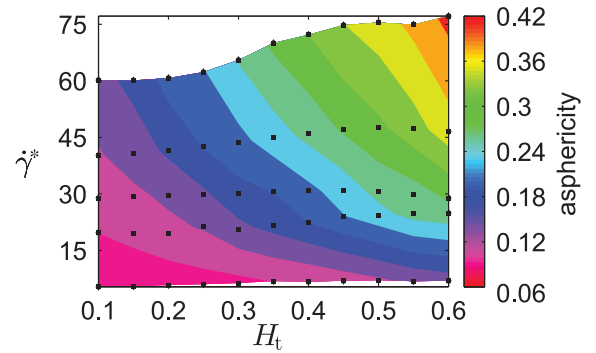


Fig. 5. Characterization of particle deformation. The asphericity α , Eq. (5), is shown for various values of hematocrit H_t and shear rate $\dot{\gamma}^*$. The color code ranges from pink (low asphericity) to red (high asphericity). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(e.g., platelets and drug carriers) would occupy the whole channel width. However, even without RBCs particle distribution may not be uniform due to shear-induced migration [34,71,72] caused by the shear-induced diffusion [73] and/or varying velocity gradients in flow. The presence of RBCs further affects particle distribution. Due to the deformability and biconcave shape of RBCs, they migrate toward a channel center. If the hematocrit is low, the RBC density is small enough allowing a considerable number of carriers to be present between the RBCs. As the hematocrit is increased, a clearly visible RBC-FL develops and the majority of carriers migrate or marginate into the RBC-FL. Available spacing between RBCs in flow affects the dependence of margination on the carrier size. Smaller carriers may fit better between the RBCs, and therefore more of such particles are able to stay within the RBC core (equivalently less number of such

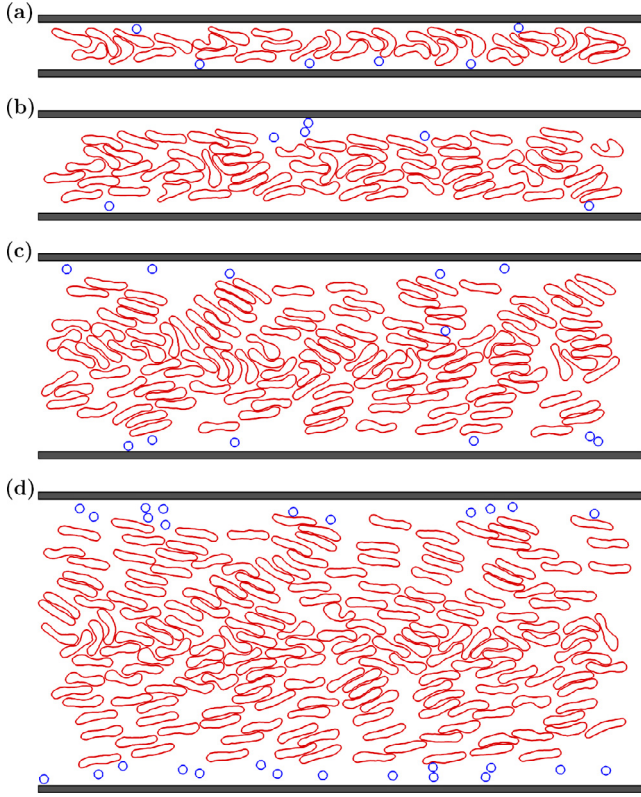


Fig. 6. Simulation snapshots in 2D for different channel widths W . The hematocrit is $H_t = 0.3$ and the shear rate $\dot{\gamma}^* = 29.7$. (a) $W = 6.1D_r$, (b) $W = 3.28D_r$, (c) $W = 6.56D_r$, and (d) $W = 9.84D_r$.

carriers would marginate into the RBC-FL) in comparison to larger particles. However, margination into the RBC-FL is not a sufficient measure to predict the adhesion potential of platelets and drug carriers. Therefore, we have focused on the probability of margination into the PAL, and we find that the thickness of the RBC-FL mainly determines the dependence of particle margination into the PAL on the carrier size. If particle size is small in comparison with the RBC-FL thickness, particles fill the full RBC-FL and thus, only a small fraction of particles is very close to a wall in the PAL. In contrast, if the particle

size is close to the RBC-FL thickness, the particle is effectively confined between the wall and the RBC core, and hence, its margination into the PAL is very pronounced. For particle sizes noticeably larger than the RBC-FL thickness, a reduction in margination may occur with an example of poor margination of WBCs at high hematocrits [37,38]. These arguments provide an explanation for the dependence of carrier margination on its size. In addition, they support the enhancement of margination into the PAL when hematocrit is increased or the shear rate is decreased, since these conditions lead to a reduction in the RBC-FL thickness. Another possibility to affect the margination of very small rigid particles is to use magnetic particles and a low magnetic field. Simulations of such a system have shown to improve particle margination due to the application of external magnetic field [74].

The dependence of particle margination on channel width is also directly associated with the RBC-FL. Generally, the thickness of RBC-FL increases with increasing characteristic vessel size, which has been also observed *in vivo* [75]. We have mainly investigated particle sizes from sub-micrometer to a few micron, which are already comparable to or smaller than a characteristic RBC-FL thickness in the smallest channel studied. In the larger channel, the RBC-FL gets even wider leading to a reduction of carrier margination into the PAL. This conclusion may change for larger particles (e.g., with a size comparable to WBCs), for which margination might be most efficient in channels of some intermediate width due to the size arguments presented above; however, it would require further investigation. In context of drug delivery, the maximum carrier size is limited by the smallest vessels (or capillaries) in human body which are about $3\text{ }\mu\text{m}$ – $4\text{ }\mu\text{m}$ in diameter, because the particles with characteristic diameters larger than about $4\text{ }\mu\text{m}$ may get trapped in the smallest capillaries [76]. This limitation might be slightly relaxed with increasing particle deformability; however, as our results show, very deformable carriers experience a stronger lift force acting away from the wall and reducing their margination. This effect might be even amplified if the relaxed shape of the particles is not spherical. An advantage of flexible particles might be at high hematocrits and low shear rates as was shown in Fig. 4. Furthermore, deformable particles are likely to have better adhesion properties than rigid carrier, since they can deform into a more flat shape increasing the area for interactions between the carrier and the wall. For example, experimental [77,78] and theoretical [79,80] studies predict that oblate ellipsoids are subject to a stronger adhesion than spheres with the same volume.

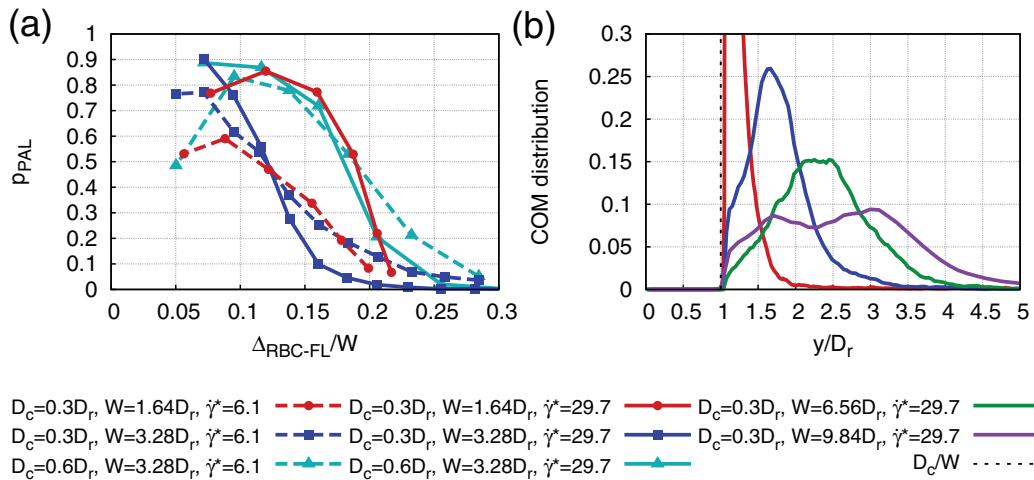


Fig. 7. Carrier distributions for different channel sizes. Comparison of a rigid carrier with a size $D_c = 0.6D_r$ in a channel of $W = 3.28D_r$ (cyan triangles) and a particle with $D_c = 0.3D_r$ in channels with $W = 1.64D_r$ (red circles), $W = 3.28D_r$ (blue squares), $W = 6.56D_r$ (cyan), and $W = 9.84D_r$ (purple). The shear rates $\dot{\gamma}^* = 6.1$ (dashed curves) and $\dot{\gamma}^* = 29.7$ (solid curves) are considered. (a) Probability p_{PAL} of margination depending on the RBC-FL thickness Δ_{RBC-FL} and scaled by the channel width W . (b) COM distribution with respect to the position y . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

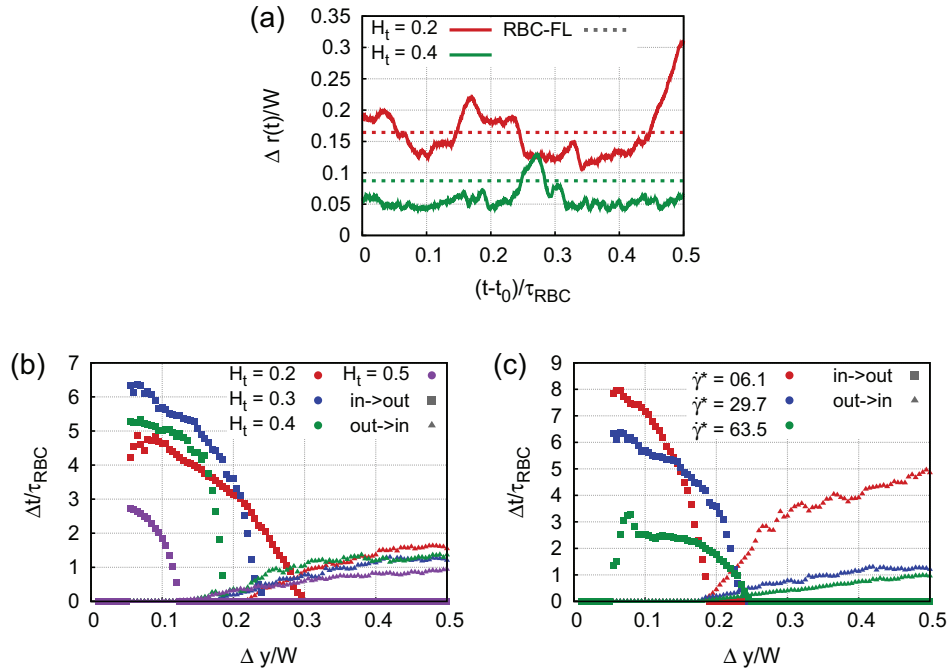


Fig. 8. Crossing of the RBC-FL border by particles. (a) Trajectory $\Delta r(t)$ of a sphere with $D_c = 0.28D_r$ normalized by the channel width W . The observation time starts from t_0 and is normalized by the RBC relaxation time τ_{RBC} . The shear rate is $\dot{\gamma}^* = 59$ and different curves correspond to two hematocrit values of $H_t = 0.2$ (red) and $H_t = 0.4$ (green). The RBC-FL border is denoted by the dashed lines. (b, c) Time interval Δt for a 2D circular carrier of size $D_c = 0.3D_r$ needed to cross the RBC-FL border depending on the COM position y . (b) $\dot{\gamma}^* = 29.7$ and different curves correspond to $H_t = 0.2$ (red), $H_t = 0.3$ (blue), $H_t = 0.4$ (green), and $H_t = 0.5$ (purple). (c) $H_t = 0.3$ and different curves correspond to $\dot{\gamma}^* = 6.1$ (red), $\dot{\gamma}^* = 29.7$ (blue), and $\dot{\gamma}^* = 63.5$ (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Another important aspect, which we already briefly mentioned, is that after particles got margined they remain quasi-trapped in the RBC-FL. Thus, it is interesting to consider the probability for a particle to escape from the RBC-FL. Fig. 8(a) shows particle trajectories for different H_t values indicating that it is much less likely for a particle to leave the RBC-FL at high hematocrits. To further analyze this behavior, Fig. 8(b) and (c) presents the time required for particles to cross the RBC-FL border with respect to their COM position in the channel for different hematocrits and shear rates. Initially the particles are located either inside or outside the RBC-FL. Particles, which were initially located inside (outside) the RBC-FL, are tracked in time until they have moved a distance of $1.15\Delta_{RBC-FL}$ ($0.85\Delta_{RBC-FL}$) to the wall, which allow the detection of RBC-FL border crossing events. Clearly, particles remain in the RBC-populated region for considerably shorter times than the times they spend in the RBC-FL. For an intermediate hematocrit value, particles remain longest in the RBC-FL, since for low hematocrit values, the particles can easily find a gap between the RBCs and leave the RBC-FL. As hematocrit is increased, the possibility to find a gap between RBCs reduces, forcing the particles to stay longer within the RBC-FL. However, for high enough hematocrit values the RBC-FL becomes small and the number of collisions between the particles and RBCs significantly increases, which may also favor crossing of the RBC-FL border by particles. These arguments also fully support the results that sub-micrometer particles marginate worse than their larger counterparts, since smaller particles have a higher probability to leave the RBC-FL. For instance, a recent theoretical analysis [81] predicts that rebound collisions between the margined platelets and RBCs play a significant role for their margination.

Besides understanding the effect of a variety of parameters on particle margination, it is also interesting to discuss the underlying margination mechanisms. The arguments above and recent theoretical and numerical studies [82] suggest that a finite size of particles influences their distribution. However, 2D Lattice-Boltzmann simulations [41] have also suggested that pure volume exclusion is

not sufficient to fully explain margination of platelets. Recent studies [34,41,45] indicate that shear-induced diffusion of particles due to the interactions with RBCs in flow is one of the main margination mechanisms. The effective radial diffusion of particles is not uniform in blood flow leading to a particle drift. The proposed mechanism is related to velocity fluctuations induced by the interactions with RBCs, which depend on the radial position of a particle in the channel and lead to a motion toward the wall [34,44,45]. Thus, an additional drift velocity toward the wall is present in such systems [34,41,45]. The other component of the margination mechanism is the lift force on RBCs and deformable particles. Furthermore, the difference in margination for soft and rigid capsules has been explained by larger displacements of rigid particles in comparison to the floppy particles through a collision of these two [33–35]. This assumption has been affirmed by a master-equation model. The difference in margination between different particle sizes is suggested to be mainly due to a different wall-induced migration (or lift force) [33–35]. Finally, the rebound collisions between the margined particles and RBCs are also of importance as discussed above. However, a full comprehensive theory of particle margination is yet to be developed.

5. Conclusion

In conclusion, we have shown that the particle margination is mainly determined by the RBC-FL thickness in relation to the particle size. Accordingly, margination increases with increasing hematocrit, decreasing shear rate, increasing particle size, and decreasing channel width, as long as the particles are not too small in comparison to the RBC-FL. The margination results indicate that a diameter of about $2\text{ }\mu\text{m}$ – $3\text{ }\mu\text{m}$ would be advantageous for drug carriers. Deformable particles seem to be advantageous for low shear rates and narrow RBC-FLs, while otherwise a rigid carrier shows better margination properties. Furthermore, the finite carrier size and collisions with RBCs determine the COM distribution of particles in the

RBC-FL, whereas hydrodynamic interactions between particles and RBCs affect the speed of margination.

Conflict of interest

The authors declare no conflict of interest.

Ethical approval

No ethical approval is required for this work.

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